Exp : 8

27/8/26

Nmap Live Host Discovery

Aim:

To discover live hosts on a network using nmap host-discovery - ARP Scans, ICMP Scan TCP, UDP ping Scan, and to understand which protocol operates at which layer of TCP/IP model.

Introduction:

Nmap (network mapper) is free-open source network scanning tool created by Gordon Lyon and first released in 1997. It is the industry standard tool for mapping networks, discovering live hosts.

this answer on 2 questions asked during reconnaissance:

(1) which systems are up?

(2) what services are running on the system

Nmap can determine whether a host is alive using 4 different protocols one at each layer of the TCP/IP Stack.

(1) ARP (Link layer) - for hosts on the same local subnet

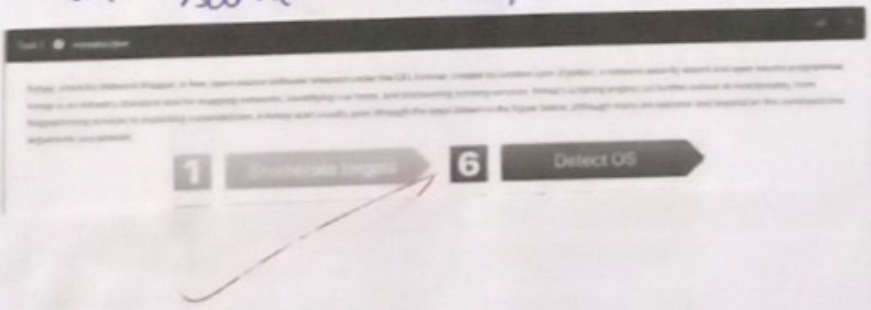
(2) ICMP (network layer) - echo, Timestamp, address mask requests

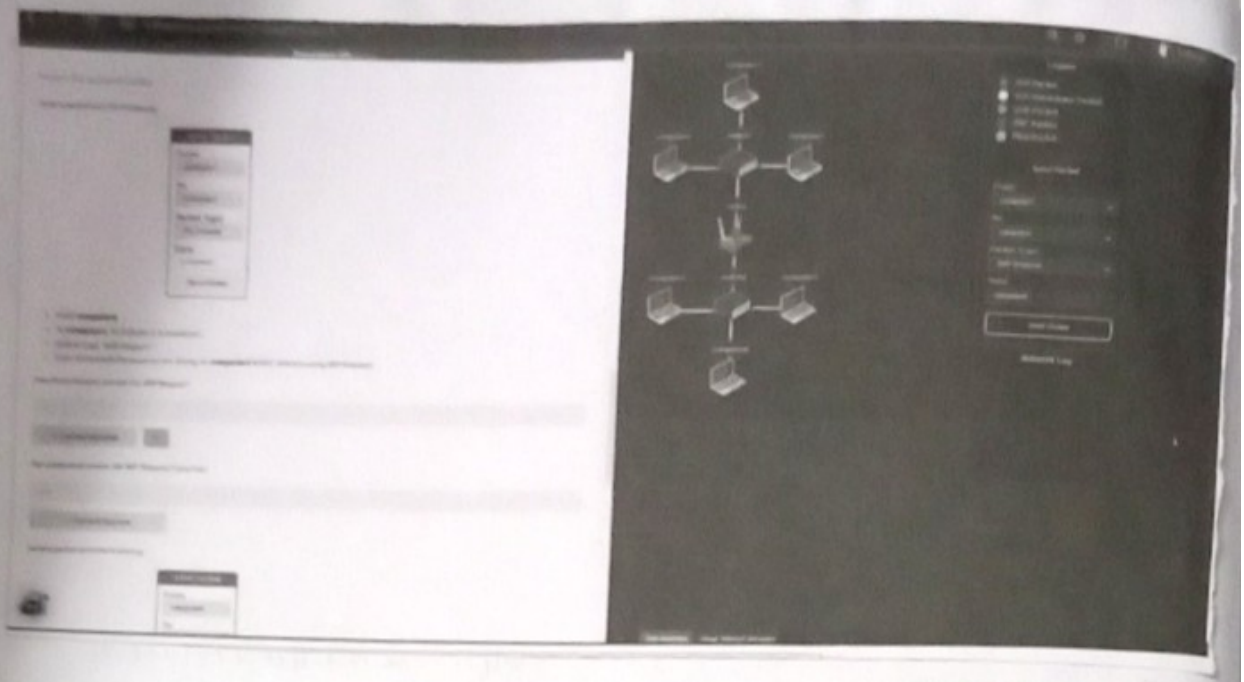
(3) TCP (transport layer) - SYN, ACK ping probes

(4) UDP (transport layer) relies on ICMP reply

(1) ARP host discovery

On a local ethernet or wifi communication requires destination's MAC address. To learn it, a host broadcast an ARP request, any host holding that IP replies with its MAC address. A reply proves that host is up. ARP only works within your own subnet, since ARP frames are not routed. By default, nmap using a local subnet automatically uses ARP. To run only an ARP based host discovery. ARP is especially useful during post exploitation internal enumeration - once inside a network it quickly reliably maps other live host on same segment.





(2) ICMP Host discovery :

ICMP offers several requests that can reveal a live host. Windows systems, many firewall block ICMP echo by default. So Nmap also supports Timestamp address - mask requests as fallback technique

7.1 ICMP Echo (-PE)

$ sudo nmap -PE ... 10.200.6.0/24

Nmap Scan report for 10.200.6.1

Host is up (0.069 latency).

Nmap Scan report for 10.200.6.50

Host is up (0.70s latency)

Nmap Scan report for 10.200.6.250

ICMP Timestam (-PP)

$nmap -PP -sn 10.200.6.0/24

Nmap Scan report for 10.200.6.50

Host is up (0.069s latency)

Nmap Scan report for 10.200.6.250

Host is up (0.67s latency)
Nmap done.

7.3 ICMP Address Mask (-PM)
most commonly filtered.

$ nmap -PM -Sn -10.200.6.0/24
Nmap done: 256 IP address Scanned in 52.14 seconds

TCP, UDP ping discovery.
when ICMP is filtered, Nmap can send crafted TCP or UDP packets, simply that a host is alive Purely from the fact responds at all

'TCP SYN Ping (-PS)
Sends a TCP packet with SYN flag To a Port. An open port SYN|Ack a closed port answer RST specify ports as PS80, P221, or PS80

$nmap -PS - Sn 10.200.6.0/24
Nmap Scan report for 10.200.6.50
Host is up (0.70s latency)
Nmap Scan report for 10.200.6.250.
Nmap done

TCP Ack Ping (-PA)
Sends a TCP packet with Ack flag set.
Because Ack does no belong to any connection
This requires a root sudo user.

$ sudo nmap -PA -sn 10.200.6.0/24

Nmap Scan report for 10.200.6.50

Host is up (0.70s latency)

Nmap Scan report for 6.200, 6.250

Host is up (0.069s latency)

NP4 ping (-PU).

Sends a UDP packet to UDP port.

$ sudo nmap $ -PU -sn 10.200.6.0/24

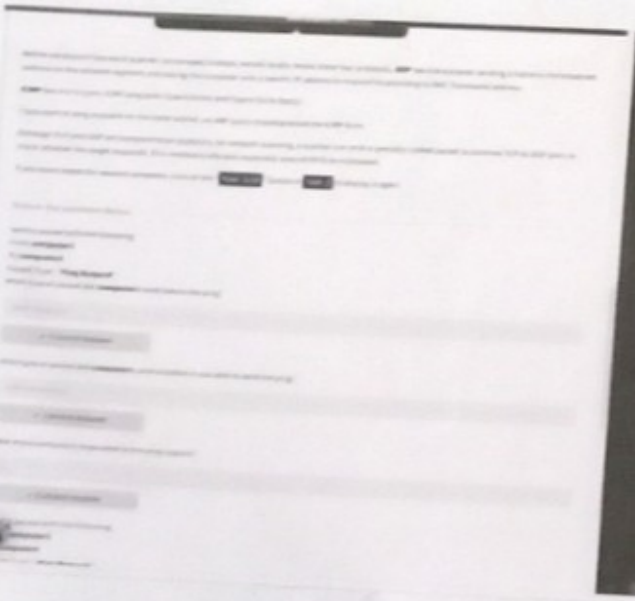
Nmap Scan report for 10.200.6.50

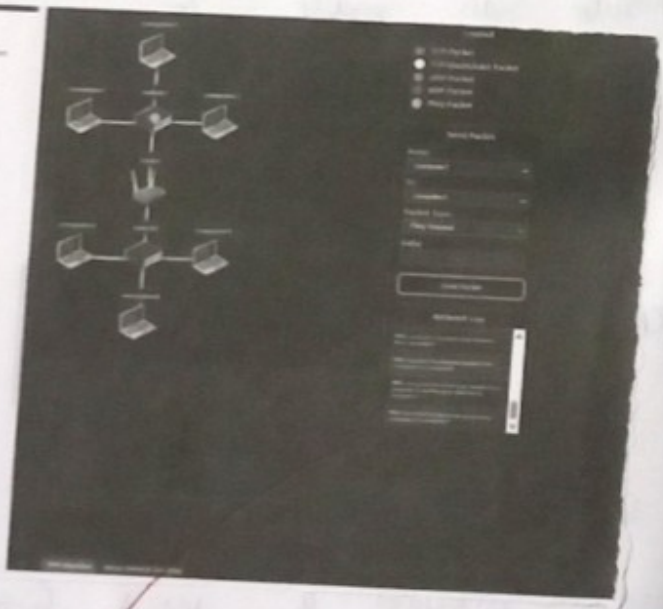
Host is up (0.70s latency)

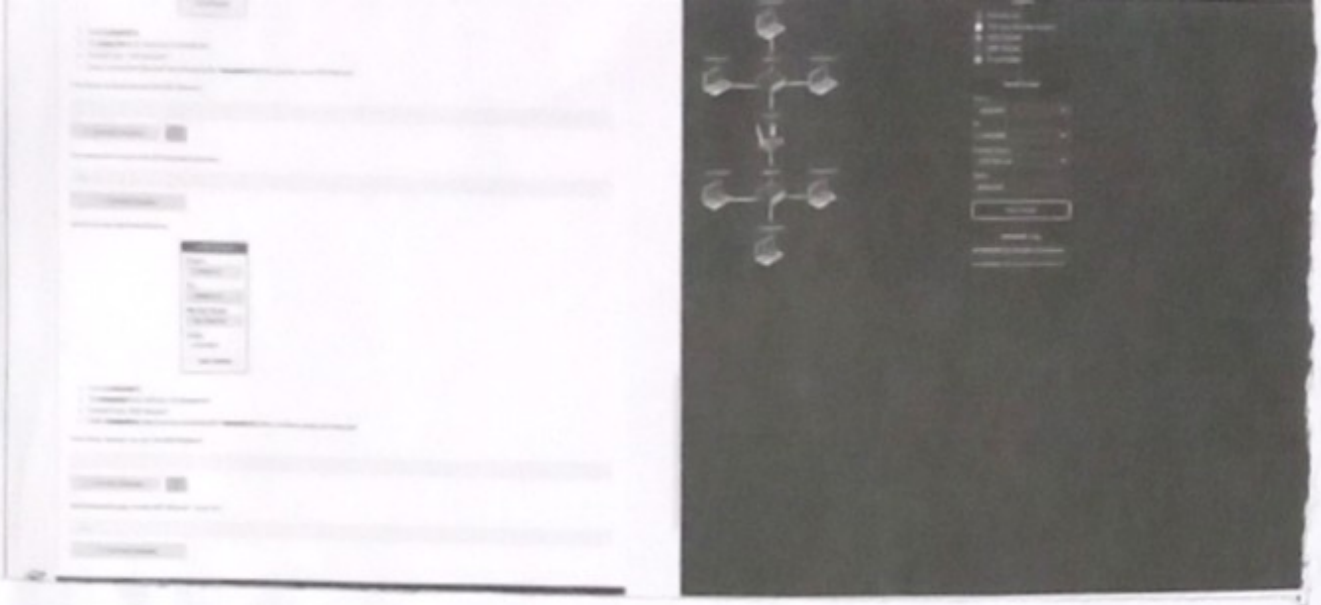
Nmap Scan report for 10.200.6.250

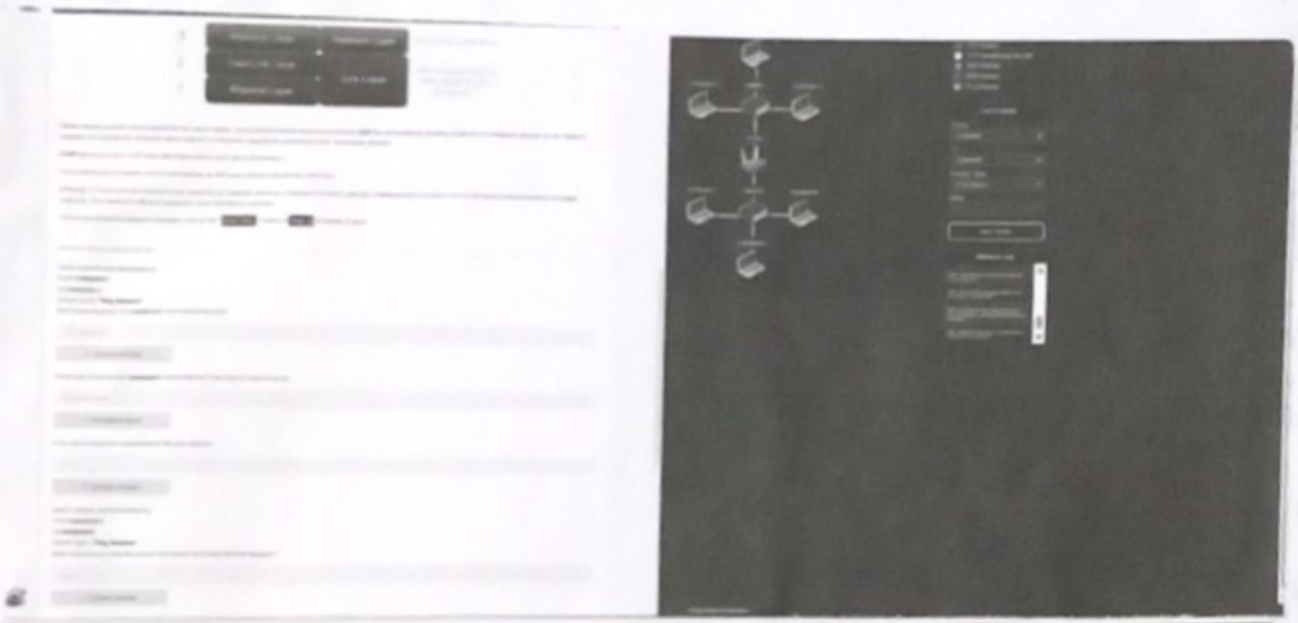
Host is up (0.069s latency)

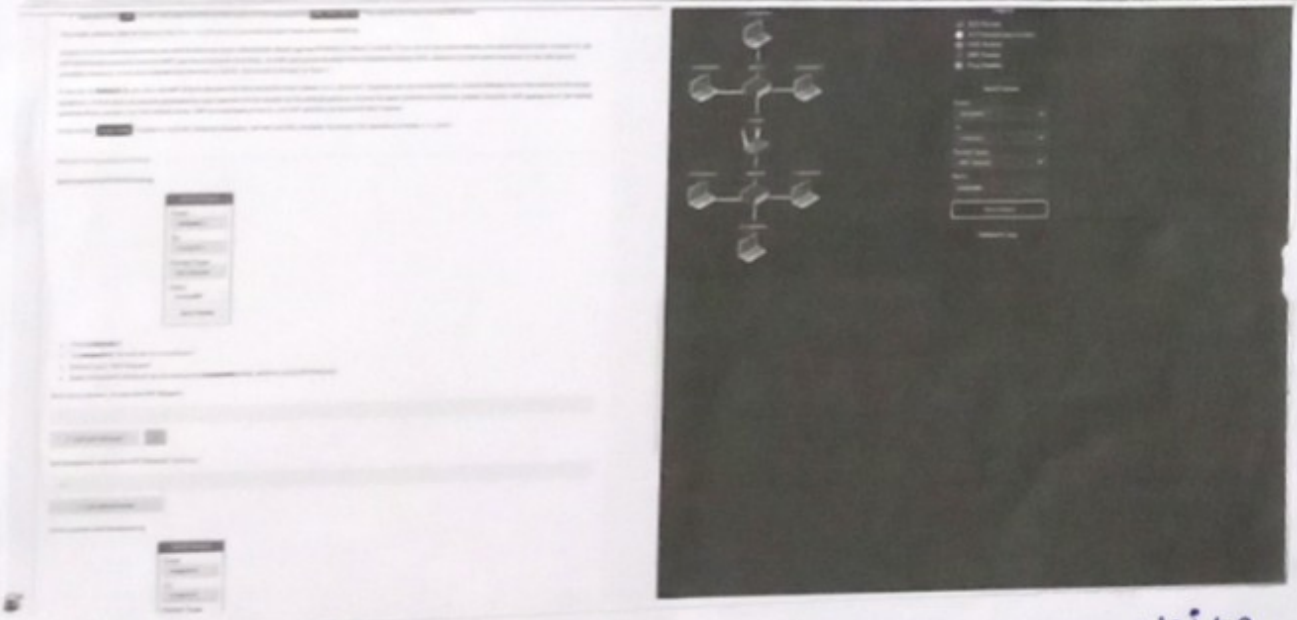
Nmap done : 256 Ip addres (2 hosts up)











Result:

Therefore Nmap Live Host Discovery using Tryhackme has been executed successfully.